

Registro de Calibración del Software (según IEC 61083-2)

1. Información de la Validación

Software Probado: Analizador de Impulsos atmosféricos tipo rayo (1.2/50 μ s). (v1.0.0) - 2026-03-24

Algoritmos soportados: Full Lightning Impulse (LI), Chopped Lightning Impulse (LIC)

Parámetros validados: Valor Pico (Ut), Tiempo de Frente (T1), Tiempo de Cola (T2), Sobrepasamiento (OS)

Generador de Datos de prueba (TDG): IEC 61083-2 TDG Version 2.1.0

Resolución TDG: 8 bits | Tasa de Muestreo: 100 MSa/s

2. Tabla de Resultados

2.1 Full Lightning Impulses - LI

Onda	Ut [kV]			T1 [μ s]			T2/Tc [μ s]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LI-A1	1049.60	1050.23	0.060%	0.840	0.840	-0.048%	60.16	60.22	0.097%	0.00	0.18	0.182%
LI-A2	1037.60	1037.76	0.016%	1.693	1.692	-0.055%	47.48	47.47	-0.028%	5.10	5.15	0.048%
LI-A3	1000.20	1001.68	0.148%	1.117	1.117	-0.038%	48.15	48.27	0.254%	4.60	4.69	0.088%
LI-A4	856.01	855.97	-0.004%	0.841	0.849	0.903%	47.80	47.83	0.065%	7.90	7.91	0.012%
LI-A5	71.97	71.98	0.015%	1.711	1.714	0.197%	47.71	47.70	-0.024%	7.70	7.76	0.063%
LI-A6	100.17	100.18	0.011%	1.762	1.767	0.276%	41.58	41.54	-0.095%	17.70	17.76	0.056%
LI-A7	104.35	104.36	0.010%	2.122	2.132	0.471%	38.36	38.39	0.082%	20.10	20.21	0.114%
LI-A8	96.01	96.02	0.012%	1.503	1.507	0.270%	44.92	44.89	-0.072%	14.80	14.78	-0.024%
LI-A9	55.93	55.93	0.007%	1.215	1.216	0.089%	55.74	55.76	0.032%	4.00	4.03	0.033%
LI-A10	81.93	81.94	0.012%	0.924	0.926	0.212%	42.66	42.67	0.019%	12.00	12.03	0.025%
LI-A11	86.60	86.60	0.001%	0.578	0.587	1.563%	56.37	56.40	0.050%	4.10	4.09	-0.009%
LI-A12	85.58	85.59	0.005%	0.587	0.588	0.198%	57.36	57.37	0.023%	2.30	2.27	-0.030%
LI-M1	952.09	952.85	0.080%	1.123	1.125	0.136%	85.60	85.70	0.117%	2.10	2.03	-0.074%
LI-M2	-1041.70	-1041.45	-0.024%	3.356	3.354	-0.070%	61.25	61.35	0.156%	9.20	9.08	-0.117%
LI-M3	-1026.50	-1026.26	-0.023%	2.150	2.147	-0.118%	41.75	41.83	0.190%	9.20	9.07	-0.127%
LI-M4	-267.14	-267.10	-0.017%	0.987	0.988	0.082%	56.22	56.18	-0.064%	4.80	4.79	-0.010%
LI-M5	-55.00	-54.98	-0.043%	2.746	2.750	0.162%	42.11	42.10	-0.016%	18.70	18.76	0.060%
LI-M6	-166.87	-166.88	0.007%	1.356	1.357	0.083%	54.74	54.73	-0.013%	3.80	3.85	0.054%
LI-M7	-1272.30	-1272.09	-0.017%	1.482	1.484	0.154%	50.03	50.09	0.122%	11.20	11.32	0.124%
LI-M8	-99.73	-99.81	0.080%	1.515	1.519	0.291%	49.36	49.27	-0.178%	-0.50	-0.36	0.136%
LI-M9	-100.04	-100.05	0.012%	0.828	0.827	-0.161%	46.65	46.74	0.203%	1.40	1.45	0.046%
LI-M10	100.26	100.27	0.005%	1.666	1.667	0.075%	60.85	60.86	0.016%	0.00	-0.10	-0.100%
LI-M11	299.32	299.31	-0.003%	1.661	1.661	-0.024%	60.95	60.84	-0.181%	-0.50	-0.33	0.172%
LI-M12	-4.32	-4.32	-0.074%	1.292	1.289	-0.271%	52.27	52.39	0.233%	-1.80	-1.83	-0.026%
LI-M13	39.46	39.53	0.175%	1.537	1.540	0.171%	46.94	46.98	0.085%	1.80	1.68	-0.118%
LI-M14	48.55	48.47	-0.171%	0.933	0.933	-0.005%	37.48	37.38	-0.277%	4.30	4.12	-0.180%
LI-M15	497.97	497.62	-0.069%	1.017	1.017	0.036%	59.19	59.18	-0.012%	-0.10	-0.14	-0.039%
LI-M16	369.21	369.42	0.058%	0.920	0.918	-0.164%	47.53	47.56	0.053%	0.80	0.89	0.089%
LI-M17	-99.35	-99.23	-0.117%	1.775	1.774	-0.040%	53.31	53.18	-0.240%	1.30	1.25	-0.049%

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2.2 Chopped Lightning Impulses (LIC)

Onda	Ut [kV]			T1 [μs]			T2/Tc [μs]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LIC-M4	0.15	0.15	0.080%	1.305	1.310	0.358%	6.00	6.01	0.134%	-0.20	0.69	0.890%
LIC-M5	-389.90	-389.54	-0.092%	0.857	0.870	1.499%	9.24	9.27	0.305%	6.80	7.04	0.243%

3. Estimación de Incertidumbre (según Anexo B.3.3)

Basado en las máximas desviaciones encontradas:

3.1 Full Lightning Impulses - LI

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.1009	0.0150	0.1020
Tiempo de Frente (T1)	0.9024	0.2500	0.9364
Tiempo de Cola (T2/Tc)	0.1597	0.0300	0.1625
Sobrepasamiento (OS)	0.1050	0.0400	0.1123

3.2 Chopped Lightning Impulses (LIC)

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.0531	0.0250	0.0587
Tiempo de Frente (T1)	0.8654	0.4500	0.9754
Tiempo de Cola (T2/Tc)	0.1763	0.1500	0.2315
Sobrepasamiento (OS)	0.5137	0.0250	0.5144